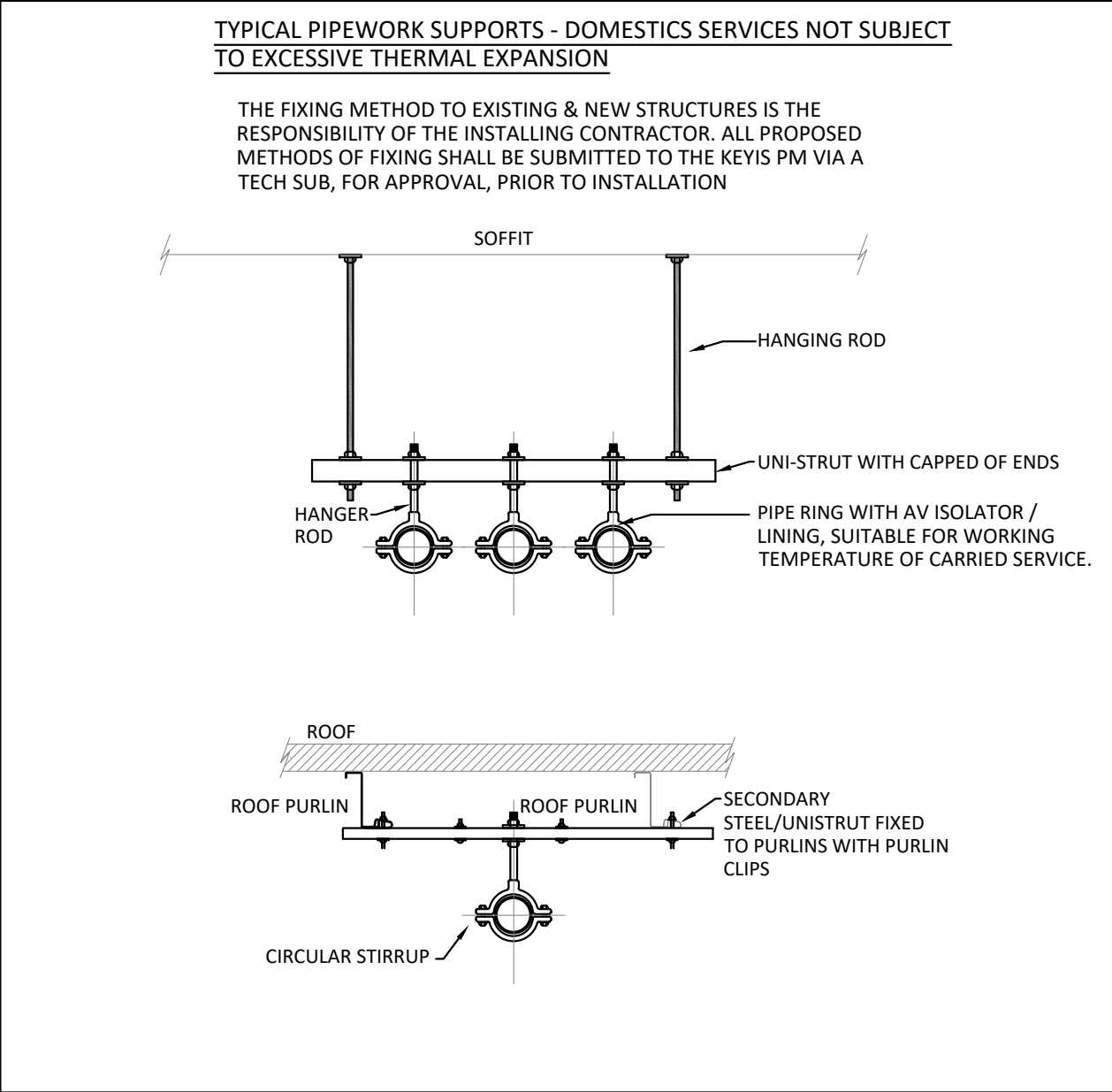




RECOMMENDED ROD SIZE					
NOMINAL PIPE DIAMETER (mm)	NOMINAL PIPE DIAMETER (in)	COPPER (in)	COPPER (mm)	STEEL (in)	STEEL (mm)
15	1/2"	3/8"	10	3/8"	10
20	3/4"	3/8"	10	3/8"	10
25	1"	3/8"	10	3/8"	10
32	1 1/2"	3/8"	10	3/8"	10
50	2"	3/8"	10	3/8"	10
65	2 1/2"	1/2"	12	1/2"	12
80	3	1/2"	12	1/2"	12
100	4	1/2"	12	5/8"	15
150	6	5/8"	15	3/4"	20
200	8	3/4"	20	3/4"	20
250	10	3/4"	20	7/8"	22
300	12	3/4"	20	7/8"	22
350	14	N/A	N/A	1"	25
400	16	N/A	N/A	1"	25
450	18	N/A	N/A	1"	25
500	20	N/A	N/A	1 1/4"	30
610	24	N/A	N/A	1 1/4"	30

COPPER/TUBE OR STAINLESS STEEL TUBES - MAXIMUM SPACE BETWEEN SUPPORT					
PIPE DIAMETER (mm)	PIPE DIAMETER (in)	HORIZONTAL RUN (m)	HORIZONTAL RUN (ft)	VERTICAL RUN (m)	VERTICAL RUN (ft)
6	N.A.	0.4	1.3	0.6	2
8	N.A.	0.6	2	0.9	3
10	3/8"	0.8	2.6	1.2	4
12	N.A.	1	3.3	1.5	5
15	1/2"	1.2	4	1.8	6
22	3/4"	1.2	4	2.4	6
28	1"	1.5	5	3	8
35	1 1/4 "	1.8	6	3	10
42	1 1/2"	1.8	6	3	10
54	2"	1.8	6	3.7	10
67	2 1/2"	2.4	8	3.7	12
76	3	2.4	8	3.7	12
108	4	2.4	8	3.7	12
133	5	3	10	3.7	12
159	6	3.7	12	3.7	12

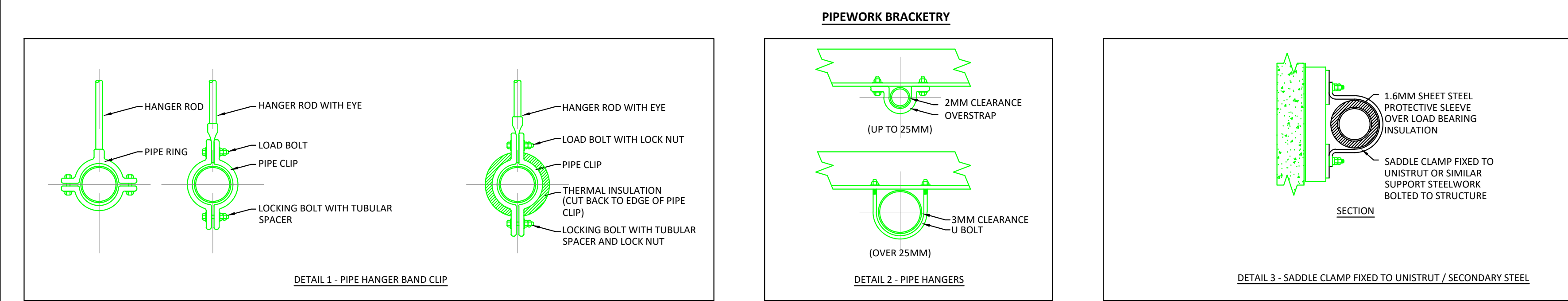
STEEL PIPE - DISTANCE BETWEEN SUPPORT		
OUTSIDE DIAMETER (mm)	HORIZONTAL RUN (m)	VERTICAL RUN (m)
15	1.5	3
20	2.4	3
25	2.4	3
32	2.4	3.7
40	2.4	3.7
50	2.4	4.6
65	3	4.6
80	3	4.6
100	3	5.5

GENERAL NOTES

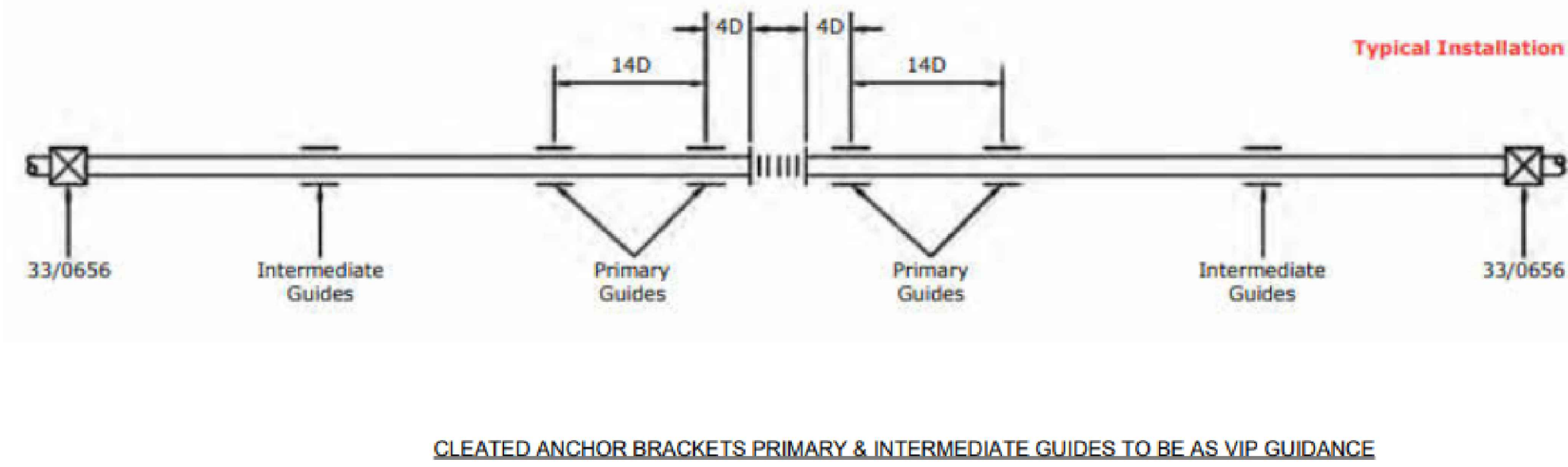
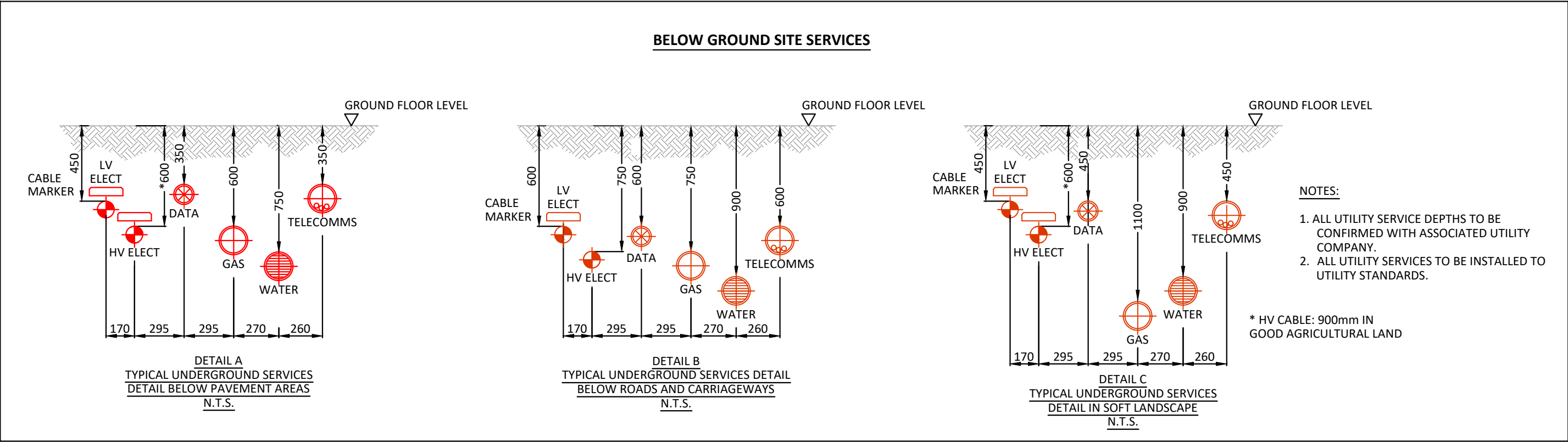
- DO NOT SCALE FROM THIS DRAWING.
- POSITIONS OF ALL FITTINGS ARE INDICATIVE ONLY.
- THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE RELEVANT SPECIFICATIONS.
- ALL FIRE STOPPING BY OTHERS, AS PART OF MAIN CONTRACTORS ELEMENT OF WORKS.
- EACH APPLIANCE SHALL BE PROVIDED WITH ANTI-VACUUM TYPE DEEP SEAL BOTTLE TRAPS.
- WHERE AIR ADMITTANCE VALVES ARE INDICATED THEY SHALL BE INSTALLED ABOVE THE APPLIANCE SPILL OVER LEVEL.
- ALL ABOVE GROUND DRAINAGE SHALL COMPLY WITH THE REQUIREMENTS OF BS EN 12056 AND BUILDING REGULATIONS APPROVED DOCUMENT PART H.
- ALL DRAINAGE PIPEWORK SHALL BE TESTED IN ACCORDANCE WITH BS EN 1610:1998.
- ALL ABOVE GROUND (EXPOSED) DRAINAGE PIPEWORK SHALL BE IDENTIFIED USING COLOUR BANDING TO BS 1710:2014.
- ALL DRAINAGE PIPEWORK PASSING THROUGH NOISE SENSITIVE AREAS SHALL BE ACOUSTICALLY SEALED AND INSULATED TO ENSURE THE ROOM ACOUSTIC PERFORMANCE IS NOT COMPROMISED.
- ALL HIGH LEVEL DRAINAGE PIPEWORK SHALL BE LAID AT A MINIMUM FALL OF 1:80 UNLESS OTHERWISE STATED.
- REFER TO ARCHITECTS SANITARY DETAILS FOR APPLIANCE CONNECTIONS.
- RODDING EYES SHALL BE PROVIDED AT CHANGES OF DIRECTION ON ALL NEW WASTE PIPE & SHALL BE FULLY ACCESSIBLE FOR MAINTENANCE PURPOSES.
- ALL NEW SVP STACKS WHICH DISCHARGE AT ROOF OR PLANTROOM ROOF LEVEL WILL TERMINATE TO ATMOSPHERE (300mm NOMINAL) ABOVE FINISHED ROOF LEVEL WITH COWL/BALLOON/CAGE TERMINAL, OR TO A DEDICATED PLENUM ONLY.
- CONDENSATE FROM AIR CONDITIONING UNITS SHALL BE INSTALLED IN UPVC & CONNECT TO THE DRAINAGE SYSTEM VIA A INLINE HEPVO TYPE TRAP.
- RAINWATER AND SOIL VENT PIPEWORK TO BE INSULATED WITHIN 3 METRES OF THE INLET CONNECTION TO PREVENT CONDENSATION FORMATION UNLESS THE PIPEWORK IN WITHIN A SERVICES RISER.

WATER SERVICES NOTES

- ALL FIRE STOPPING BY OTHERS, AS PART OF MAIN CONTRACTORS ELEMENT OF WORKS.
- ALL CONCEALED HWS PIPEWORK SHALL BE INSULATED IN ACCORDANCE WITH THE PROJECT SPECIFICATION.
- ALL BCWS/MCWS PIPEWORK SHALL BE INSULATED AND VAPOUR SEALED IN ACCORDANCE WITH THE PROJECT SPECIFICATION.
- ALL VISIBLE PIPEWORK AND FITTINGS (EXCLUDING VALVES AND CHROME FINISH) SHALL BE LEFT READY FOR DECORATIVE PAINTING BY OTHERS.
- ALL INDIVIDUAL BRANCH PIPES TO ITEMS OF SANITARY WARE SHALL BE COMPLETE WITH BALL-O-FIX TYPE ISOLATION VALVES.
- ALL PIPEWORK DROPS TO BE CONTAINED IN IPS PANELS/STUD PARTITIONING TO BE INSULATED.
- AIR VENTS TO BE FITTED AT ALL HIGH POINTS IN SYSTEM PIPEWORK.
- DRAIN COCKS SHALL BE FITTED AT LOW POINTS IN SYSTEM PIPEWORK.
- ALL SERVICES DROPPING TO LOW LEVEL TO BE IN WALLIPS IF POSSIBLE. BOXING OUT TO BE PROVIDED (BY OTHERS) WHERE PIPEWORK CAN NOT BE CONCEALED.
- ALLOW ALL DROPS, BRANCHES & DIRECTION CHANGES TO BE FREE TO FLEX.



PLEASE NOTE THAT THE DETAILS SHOWN ABOVE ARE COMMON DETAILS, BUT NOT EXHAUSTIVE. IT IS THE RESPONSIBILITY OF THE INSTALLING CONTRACTOR TO ENSURE THAT THE BRACKET DETAIL AND SUPPORTS ARE SUITABLE FOR THE MEDIUM/PIPEWORK TYPE ETC, INCLUDING ANY SECONDARY SUPPORTS BETWEEN STRUCTURAL STEEL AND PIPEWORK. ANY CALCULATION REQUIRED TO PROVE SUPPORT DETAILS ARE SUITABLE FOR THE REQUIRED THRUST LOADS ETC ARE TO BE PROVIDED BY A SUITABLY QUALIFIED PERSON I.E. STRUCTURAL ENGINEER OR SUPPORT FRAMING PROVIDER.



PLEASE NOTE THAT THE DETAILS SHOWN HERE ARE BASED ON VIP AS AN EXAMPLE ONLY. THE POSITION OF THE GUIDES ETC SHOULD BE AS RECOMMENDED BY THE MANUFACTURER PROVIDING THE ANCHORS/GUIDES. IT IS THE RESPONSIBILITY OF THE INSTALLING CONTRACTOR TO ENSURE THAT THE BRACKET DETAIL AND SUPPORTS ARE SUITABLE FOR THE MEDIUM/PIPEWORK TYPE ETC, INCLUDING ANY SECONDARY SUPPORTS BETWEEN STRUCTURAL STEEL AND PIPEWORK. ANY CALCULATION REQUIRED TO PROVE SUPPORT DETAILS ARE SUITABLE FOR THE REQUIRED THRUST LOADS ETC ARE TO BE PROVIDED BY A SUITABLY QUALIFIED PERSON I.E. STRUCTURAL ENGINEER OR SUPPORT FRAMING PROVIDER.

PIPEWORK WEIGHTS

Service	Nominal Size	Material	Weight of Pipe Dry	Pipe Volume	Total Pipe Weight Wet	Insulation Density	Insulation Thickness	Insulation Weight	Total Weight of pipe and insulation
	mm		kg/m	Litres/m	kg/m	kg/m3	(mm)	kg/m	kg/m
Chilled Water	50	Steel	7.404	2.07	9.47	35	50	0.27	9.75
LTHW	65	Steel	9.48	3.53	13.01	35	50	0.36	13.37
	80	Steel	12.12	4.91	17.03	35	50	0.44	17.46
	100	Steel	17.28	8.38	25.66	35	50	0.55	26.21
	150	Steel	25.44	18.70	44.14	35	50	0.82	44.96
Chilled Water	15	S/S Mpress	0.504	0.20	0.71	35	50	0.08	0.79
LTHW	20	S/S Mpress	0.744	0.30	1.05	35	50	0.11	1.16
	25	S/S Mpress	0.96	0.51	1.47	35	50	0.14	1.61
	32	S/S Mpress	1.512	0.80	2.31	35	50	0.18	2.49
	40	S/S Mpress	1.824	1.19	3.02	35	50	0.22	3.24
	50	S/S Mpress	3.156	2.04	5.20	35	50	0.27	5.47
	65	S/S Mpress	4.452	4.08	8.54	35	50	0.36	8.99
	80	S/S Mpress	5.22	5.66	10.88	35	50	0.44	11.32
	100	S/S Mpress	6.372	8.50	14.87	35	50	0.55	15.42
Steam	50	Steel	7.404	2.07	9.47	220	80	2.76	10.18
	65	Steel	9.48	3.53	13.01	220	80	3.59	13.09
	80	Steel	12.12	4.91	17.03	220	80	4.42	16.56
	100	Steel	17.28	8.38	25.66	220	80	5.53	22.84
	150	Steel	25.44	18.70	44.14	220	80	8.29	33.81
Condensate	15	Steel	1.74	0.19	1.93	220	80	0.83	2.76
	20	Steel	2.28	0.33	2.61	220	80	1.11	3.71
	25	Steel	3.564	0.52	4.08	220	80	1.38	5.46
	32	Steel	4.608	0.93	5.53	220	80	1.77	7.30
	40	Steel	5.316	1.27	6.59	220	80	2.21	8.80
BCW	15	Copper	0.336	0.15	0.48	35	50	0.08	0.56
	22	Copper	0.624	0.32	0.95	35	50	0.12	1.07
	28	Copper	0.816	0.54	1.36	35	50	0.15	1.51
	35	Copper	1.344	0.84	2.18	35	50	0.19	2.37
	42	Copper	1.632	1.23	2.87	35	50	0.23	3.10
	54	Copper	2.112	2.10	4.21	35	50	0.30	4.50
	65	304 dairy	4.416	3.32	7.73	35	50	0.36	8.09
	80	304 dairy	5.412	5.03	10.44	35	50	0.44	10.88
	100	304 dairy	6.996	7.85	14.85	35	50	0.55	15.40
DWS	15	Copper	0.336	0.15	0.48	35	50	0.08	0.56
	22	Copper	0.624	0.32	0.95	35	50	0.12	1.07
	28	Copper	0.816	0.54	1.36	35	50	0.15	1.51
	35	Copper	1.344	0.84	2.18	35	50	0.19	2.37
	42	Copper	1.632	1.23	2.87	35	50	0.23	3.10
	54	Copper	2.112	2.10	4.21	35	50	0.30	4.50
Compressed Air	15	Thin Wall Galv steel	0.81	N/A	0.81				0.81
	22	Thin Wall Galv steel	0.99	N/A	0.99				0.99
	28	Thin Wall Galv steel	1.51	N/A	1.51				1.51
	35	Thin Wall Galv steel	1.95	N/A	1.95				1.95
	42	Thin Wall Galv steel	2.47	N/A	2.47				2.47
	54	Thin Wall Galv steel	2.99	N/A	2.99				2.99
	65	304 dairy	4.416	N/A	4.42				4.42
	80	304 dairy	5.412	N/A	5.41				5.41
	100	304 dairy	6.996	N/A	7.00				7.00

Please note that insulation thicknesses of 50mm and 80mm and density of 35 & 220kg/m3 have been used for simplicity. Actual insulation thickness/density to be inline with British Standards Requirements

NOTES:
1. THIS DRAWING IS THE COPYRIGHTED OF KEY IS. IT MUST NOT BE REPRODUCED OR USED FOR ANY PURPOSE WITHOUT PRIOR CONSENT.
2. DO NOT SCALE FROM THIS DRAWING. WORK TO FIGURED DIMENSIONS ONLY. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS STATED OTHERWISE.
3. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH ALL OTHER MECHANICAL, ELECTRICAL & PUBLIC HEALTH DRAWING, SPECIFICATIONS AND SCHEDULES.

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BS1192 Document Number		RIBA Stage
Status		Revision